

Introduction

Diabetes is one of the most common chronic health conditions worldwide and affects millions of people each year. Relational databases provide an efficient way to organize and analyze large amounts of medical information. The purpose of this project was to design and implement a relational database focused on diabetes patient analysis using a real-world diabetes dataset containing demographic, lifestyle, medical history, genetics, organ function, and diagnosis information. Our database was designed to transform a large flat dataset into a normalized relational structure that could better support efficient storage and analysis of patient-related information.

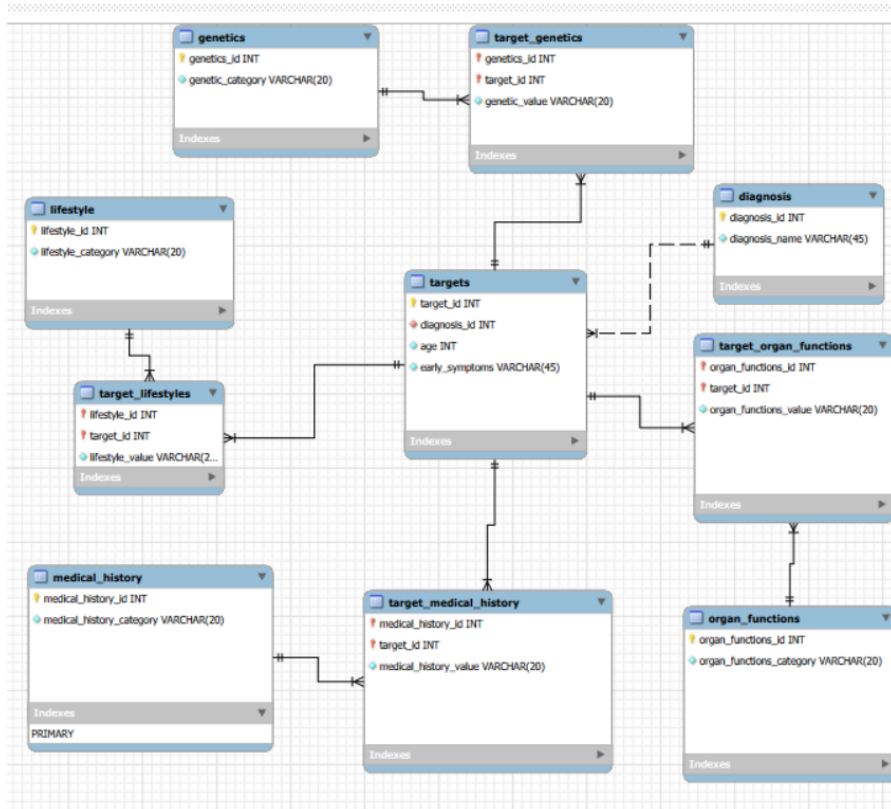
The final database consists of multiple related tables connected through primary keys, foreign keys, and linking tables that model many-to-many relationships between targets and various health-related categories. By organizing the data into a structured relational schema, the database allows users to perform SQL queries and generate insights related to diabetes diagnoses, risk factors, symptoms, and lifestyle patterns. Throughout the development process, our team applied concepts related to logical database design, physical database implementation, normalization, sample data management, and SQL query development. This report discusses the design and implementation of the database, the challenges encountered during development, the lessons learned throughout the project, and potential future improvements to the system.

Physical Database

Our database contains ten tables with targets being the main table. The targets table has a one to many relationship with four other tables; the target_genetics, target_organ_functions, target_lifestyles, and target_medical_history tables. These four tables are all linking tables, and have a many to one relationship with its corresponding tables; genetics, organ_functions, lifestyle, and medical_history tables. This means that the targets table has a many to many relationship with genetics, organ_functions, lifestyle, and medical_history. The final table,

diagnosis, has a one to many relationship with targets, as many targets can share the same diagnosis.

Logical Design



Our database is designed so that the targets table represents the core of our schema. Each record links to all linking tables which house categorical data, allowing researchers to access only relevant data based on their query needs. This also makes the database easier to scale, as new data can be added to each table separately using just the target_id.

Sample Data

The sample data used in our database was derived from the original diabetes dataset from kaggle, containing approximately 70,000 target records, which for the purposes of this project, we decided to insert a representative subset of the data into our MySQL database while preserving the relational structure of the schema. We filtered the dataset to only the most common diabetes, and then reduced that dataset to the first 100 for the database's readability. The general data is correlated with multiple genetics, lifestyle, organ function, and medical

history categories, which are connected through many-to-many relationships with each target from the targets table.

The sample data was cleaned and organized to maintain consistency across the database. For example, diagnosis categories, lifestyle classifications, and medical history values were standardized before insertion into the tables. This helped maintain the referential integrity. More importantly it allows the linking tables and foreign key constraints to function correctly. The inserted sample data allowed us to effectively test joins and aggregations, along with filtering operations, and finally subqueries through the views included in the database. By using realistic data regarding different types of diabetes, we were able to simulate how researchers and healthcare analysts may query patient information to identify trends and risk factors.

Here's the dataset of the genetics table, and the first 10 rows of the target_genetics table for reference of how the data correlates:

genetics_id	genetics_category	genetic_id	target_id	genetic_value
	1 genetic_markers	1	1	Positive
	2 family_history	1	2	Positive
	3 ethnicity	1	3	Negative
	4 genetic_testing	1	4	Negative
		1	5	Positive
		1	6	Positive
		1	7	Positive
		1	8	Negative
		1	9	Positive
		1	10	Positive

Views and Queries

The first view that we created was view_targets_genetic_marker. This allowed us to collect each target's age, early symptoms, diagnosis, and genetic marker result. It joins the targets, diagnosis, target_genetics, and genetics tables to filter for genetic markers. This is useful for researchers interested in seeing how genetic markers relate to different diabetes diagnoses. Our next view was view_diagnosis_stats, which counts the number of targets per diagnosis and calculates their average age. It filters out diagnosis groups where the average age is 25 or under by using a HAVING clause, helping focus on adult-onset diagnosis patterns. We then have view_symptomatic_high_risk, which identifies targets who exhibit early symptoms while also carrying a high-risk ethnicity classification. This view highlights a possible high-risk group of targets that may need more attention. Next is view_positive_genetic_testing, which calculates the

number of targets for each diagnosis that had a positive genetic testing result. This gives information about the diagnosis groups that had more cases of positive genetic testing throughout the dataset. The fifth view, `view_older_than_avg`, utilizes a correlated subquery to extract all targets whose age is above the average age of others with the same diagnosis. This helps identify outliers within diagnosis age groups. Lastly, our sixth view is `view_lifestyles_by_diagnosis`. This links the `target_lifestyles` linking table to its two source tables and gives information about the number of targets with particular lifestyles for each diagnosis type.

View Name	Req. A	Req. B	Req. C	Req. D	Req. E
<code>view_targets_genetic_markers</code>	X	X		X	
<code>view_diagnosis_stats</code>	X	X	X		
<code>view_symptomatic_high_risk</code>	X	X			
<code>view_positive_genetic_testing</code>	X	X	X		
<code>view_older_than_avg</code>	X				X
<code>view_lifestyle_by_diagnosis</code>	X		X	X	

Changes from Original Design

Throughout the development process of the database, we made several crucial changes to the original design. Our main goals were to reduce redundancy, and also improve normalization. In the original design one of our major issues was that our ERD had a simpler structure, containing primarily one to one relationships between the tables. After receiving feedback, we redesigned the schema to support more many to many relationships, specifically between targets

and the genetics, lifestyle, and organ function(with the use of linking tables). This allowed the database to have improved flexibility and scalability, and also allowed us to make queries more efficient and reduce redundancy such as duplicated data. Another step we took to improve the database was we refined the scope by removing attributes and categories that we decided were less relevant analyzing diabetes, which was the primary focus. An example is putting clinical tests and pregnancy information into broader categories such as medical history and organ function.

Database Ethics

Handling sensitive patient data, even for research purposes in academic projects, comes with several important ethical considerations. The information in our data set included health details from 70,000 patients with very sensitive information. Although the data set utilized in our study was public, it would still be necessary to emphasize that any implementation of such databases in the real world would involve abiding by certain regulations that stipulate how patient health data should be managed, accessed, and transferred; this includes adhering to HIPAA guidelines. Patient consent would be necessary to collect this information, and the identities of the individuals whose health records were accessed should always remain confidential.

There is also a potential for biased diabetes datasets that need to be considered from an ethical perspective. Any results obtained by queries and analyzed will further perpetuate any health inequality based on how the dataset represents certain demographics. For example, in the event that our database favored either age or ethnicity, results may not apply equally to other groups in society. The use of the database responsibly requires knowing its origins, limitations, and how it may skew towards one particular group of people.

Lastly, our consideration to only include the most common forms of diabetes is convenient but unethical. People diagnosed with rare diabetes will be left out and will not benefit from any insights made from the database. A possible improvement we can do in the next project is by including all forms of diabetes – even if rare – to help a more diverse population of patients.

Lessons Learned

A valuable lesson we learned through this experience was how difficult it is to create a normalized database out of a flat database. Our original ERD had a more simplistic model with a greater amount of one-to-one relationships among the tables. After several iterations of feedback, we found out that such an arrangement would lead to data redundancy and make queries based on combinations of patients' properties more complicated. The redesigning process when we created many-to-many relationships of targets with genetics, lifestyle, organs functions, and medical history through linking tables was very difficult but very rewarding.

Dealing with actual health records has shown us that any raw database will always need some processing before it can be placed into a normalized form. The task of selecting relevant columns, overlapping columns, and deciding what data should go into what table is quite complicated and requires some reasoning skills as well. As an example, clinical tests and pregnancy history have been at first considered separately from each other, but after the whole process, we decided that it was more appropriate to consider them as medical history and organ functions correspondingly.

Writing the SQL views and queries required us to think about what kind of question a researcher or physician would want to answer from a database. While it is fairly simple to write technically correct queries, it is much more difficult to craft an efficient and functional query. The importance of aliasing, use of filters in WHERE clauses, and table joins that do not introduce performance problems became apparent during this exercise. The creation of six views helped us understand the role views play in the process of abstraction of a complex database.

Potential Future Work

To update our current database schema and bring improvements to the scope and analysis of this database, we aim to include additional forms of diabetes that aren't as widely talked about as the seven most common types. Among the types of diabetes we have included thus far, future updates to our schema may include neonatal diabetes, Wolfram syndrome, latent autoimmune diabetes in adults or LADA, Type 3c diabetes, monogenic diabetes, and Type 5 diabetes to strengthen the scope of our dataset (Diabetes UK).

In other areas of our database, many optimizations and constraints can be added so that our database runs more efficiently. For example, many fields from our original dataset resulted in Yes/No answers that could be stored as binary formats such as using int data type formats to

show 0 (False) or 1 (True). Furthermore, given that future database updates may use other datasets, the ability to check when a target's medical history has been created or updated through TIMESTAMP columns would improve the reliability of our data.

References

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